

Diagnostic Summary Report

Patient Overview

- **Patient Name:** Ashley Grahm
- **Age:** 20
- **Gender:** Female
- **Primary Complaint:** Headache, fever

Red-Flag Alert

- No critical red flags detected.

Risk Stratification

- **Level:** Medium
- **Key Drivers:**
 - Fasting Blood Sugar = 141 mg/dL (high)
 - HbA1c = 7.1 % (diabetic range)
 - Frequent urination, nausea, weakness reported by patient
 - Temperature = 100.4 °F and WBC = 10,570/μL (mild leukocytosis)
 - Absence of severe vital sign abnormalities (BP 124/78, Pulse 70, SpO₂ 98 %)

Probable Diagnosis

- **Condition:** New-onset type 2 diabetes mellitus presenting with hyperglycemia (fasting glucose 141 mg/dL, HbA1c 7.1 %) and associated polyuria, nausea, and weakness, likely accompanied by a mild febrile illness.
- **Confidence:** 78 %
- **Justification (Explainability Pack):**
- **Laboratory evidence:** Elevated fasting glucose and HbA1c meet diagnostic criteria for diabetes.

- **Patient-reported symptoms:** Frequent urination, nausea, weakness, and headache align with hyperglycemia-related dehydration and infection.
- **Vital signs:** Low-grade fever (100.4 °F) with mild leukocytosis supports a concurrent mild infection but does not indicate severe sepsis or DKA.

Differential Diagnoses

- Urinary tract infection (UTI) – could explain fever, polyuria, and leukocytosis (no dysuria reported).
- Viral upper respiratory infection – mild fever and leukocytosis possible, but no respiratory symptoms.
- Vitamin B12 deficiency–related anemia/neuropathy – low B12 and high homocysteine present; acute fever unlikely.
- Early meningitis – headache and fever raise concern, but no neck stiffness or photophobia.
- Pregnancy-related hyperglycemia – female of childbearing age; pregnancy status unknown.

Key Laboratory Findings

- **High WBC (10,570/ μ L)** – mild leukocytosis.
- **Fasting Blood Sugar 141 mg/dL** – hyperglycemia.
- **HbA1c 7.1 %** – diabetic range.
- **Vitamin B12 < 148 pg/mL** – deficiency.
- **Homocysteine 23.86 μ mol/L** – elevated, indicating cardiovascular risk.
- **IgE 492.30 IU/mL** – markedly elevated (possible atopy/allergy).
- **25-OH Vitamin D 8.98 ng/mL** – severe deficiency.
- **Triglycerides 168 mg/dL** – elevated.
- **Direct LDL 100.39 mg/dL** – slightly above target.

Recommended Plan

Required Investigations

- **Urinalysis with glucose and ketones** – screens for glucosuria/ketonuria and infection.

- **Urine culture** – confirms/excludes UTI.
- **Basic metabolic panel (electrolytes, BUN, creatinine)** – assesses renal function and electrolyte status.
- **CBC with differential (repeat)** – monitors WBC trend and anemia.
- **CRP and ESR** – gauges inflammatory burden.
- **Blood cultures** (if fever persists >48 h or clinical deterioration) – rule out bacteremia.
- **Oral glucose tolerance test** (if diabetes confirmation needed) – clarifies glucose tolerance status.
- **Thyroid function tests (TSH, free T4)** – excludes hyperthyroidism as a cause of weakness.
- **Pregnancy test (urine hCG)** – essential before initiating many medications.
- **Neurologic exam and, if indicated, nerve conduction studies** – evaluate possible B12-related neuropathy.
- **Repeat vitamin B12 level after supplementation** – confirms repletion.
- **Chest X-ray** (if respiratory symptoms develop) – screens for pulmonary infection.

Suggested Medications / Treatments

- **Metformin 500 mg PO daily** – first-line for newly diagnosed type 2 diabetes (titrate as tolerated).
- **Cyanocobalamin 1000 µg IM weekly × 4 weeks, then monthly** – corrects vitamin B12 deficiency (or high-dose oral B12).
- **Vitamin D3 (cholecalciferol) 2000 IU PO daily** – addresses severe vitamin D deficiency.
- **Low-intensity statin (e.g., rosuvastatin 5 mg PO daily)** – for elevated LDL/triglycerides after full cardiovascular risk assessment.
- **Acetaminophen 500 mg PO q6h PRN** – for fever and headache relief.
- **Ondansetron 4 mg PO q8h PRN** – for persistent nausea.

Medication Disclaimer: A qualified human doctor must make the final prescribing decision.

Information Gaps

- Urinalysis and urine culture results to confirm or exclude a urinary tract infection.
- Pregnancy status, which influences medication choices.
- Detailed physical exam findings (e.g., neck stiffness, neurologic deficits).
- Dietary and lifestyle information to guide diabetes education.

- Duration and progression of fever beyond the initial hour.